SCOVILLE SPLICE

SOP-AN-001
Capsaicinoid Quantification and Scoville Heat Unit Determination by Reversed-Phase High-Performance Liquid Chromatography

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ANALYTICAL METHOD

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1 Purpose and Scope

1.1 Purpose

This Standard Operating Procedure establishes the validated analytical method for the quantitative determination of individual capsaicinoids—capsaicin (CAP), dihydrocapsaicin (DHC), nordihydrocapsaicin (NDC), homocapsaicin (HC), and homodihydrocapsaicin (HDC)—in all Scoville Splice finished-product formulations. The method enables calculation of total capsaicinoid content (mg/g dry weight) and the corresponding Scoville Heat Unit (SHU) rating for each product in the portfolio.

1.2 Scope

The method applies to all ten (10) products in the current Scoville Splice lineup, spanning a capsaicinoid concentration range of approximately 220–878 mg/g DW and a corresponding pungency range of 3.0×10^{6}–13.0×10^{6} SHU. This range demands rigorous analytical precision, as the products occupy the extreme upper end of the capsaicinoid spectrum well above pure oleoresin capsicum benchmarks.

The method is applicable to:

- Raw material incoming inspection of capsaicinoid concentrates
- In-process quality control during blending and formulation
- Finished-product lot release testing
- Stability studies per ICH Q1A(R2) protocols
- Out-of-specification (OOS) and out-of-trend (OOT) investigations

2 Reference Methods and Normative Documents

Designation	Title	Application
AOAC 995.03	Capsaicinoids in Capsicums and Their Extractives—Liquid Chromatographic Method	Primary extraction and chromatographic reference
ASTA 21.3	Pungency of Capsicums and Their Oleoresins (HPLC Method)	SHU conversion factors and pungency classification
ISO 7543-2:1993	Chillies and Chilli Oleoresins—Determination of Total Capsaicinoid Content, Part 2: HPLC Method	International harmonization of extraction solvent system
ICH Q2(R1)	Validation of Analytical Procedures: Text and Methodology	Method validation parameters and acceptance criteria
ICH Q2(R2)/Q14	Analytical Procedure Development and Revision of Q2(R1)	Enhanced analytical lifecycle management approach
EURACHEM/CITAC	Quantifying Uncertainty in Analytical Measurement (3rd ed.)	Measurement uncertainty estimation framework
USP <621>	Chromatography—System Suitability	System suitability testing criteria

3 Definitions and Abbreviations

Abbreviation	Definition
CAP	Capsaicin (8-methyl-*N*-vanillyl-6-nonenamide); CAS 404-86-4
DHC	Dihydrocapsaicin (8-methyl-*N*-vanillylnonanamide); CAS 19408-84-5
NDC	Nordihydrocapsaicin (*N*-vanillyloctanamide); CAS 28789-35-7
HC	Homocapsaicin; CAS 58493-48-4
HDC	Homodihydrocapsaicin; CAS 20279-06-5
SHU	Scoville Heat Units—organoleptic pungency measure
DW	Dry weight
RP-HPLC	Reversed-phase high-performance liquid chromatography
ACN	Acetonitrile, HPLC grade
MeOH	Methanol, HPLC grade
AcOH	Acetic acid, glacial, ACS grade
CRM	Certified Reference Material
LOD	Limit of Detection
LOQ	Limit of Quantitation
RSD	Relative Standard Deviation
SST	System Suitability Test
OOS	Out of Specification

4 Equipment and Instrumentation

4.1 HPLC System

Component	Specification
HPLC System	Quaternary gradient pump with degasser, autosampler, column oven, UV-Vis detector
Analytical Column	C18 reversed-phase, 250 mm × 4.6 mm I.D., 5 µm particle size (e.g., Agilent Zorbax Eclipse XDB-C18 or equivalent)
Guard Column	C18 guard cartridge, 4.0 mm × 3.0 mm, matched stationary phase
Detector Wavelength	UV 280 nm
Injection Volume	20 µL (fixed loop or partial-loop with overfill)
Data System	Chromatography data system (CDS) compliant with 21 CFR Part 11

4.2 Auxiliary Equipment

Item	Specification
Analytical balance	Readability 0.1 mg, calibrated per ISO 8655
Drying oven	Forced-air convection, 60 ± 2 °C, verified with calibrated thermometer
Laboratory mill	Capable of producing particle size < 0.5 mm (e.g., IKA A11)
Vortex mixer	Variable speed, 0–3000 rpm
Ultrasonic bath	40 kHz, temperature controlled to 40 ± 2 °C
Centrifuge	Capable of 5000 × *g*, accommodating 50 mL tubes
Nitrogen evaporator	Controlled to 40 ± 2 °C under N_2 stream
Volumetric flasks	Class A, 10 mL, 25 mL, 50 mL, 100 mL
Micropipettes	Calibrated, 10–100 µL and 100–1000 µL
Syringe filters	PTFE, 0.45 µm, 13 mm diameter

5 Reagents, Standards, and Reference Materials

5.1 Solvents

- Acetonitrile (ACN), HPLC gradient grade, ≥99.9%
- Methanol (MeOH), HPLC grade, ≥99.9%
- Acetone, ACS reagent grade, ≥99.5%
- Acetic acid, glacial, ACS grade, ≥99.7%
- Water, Type I (18.2 MΩ·cm), freshly dispensed
- Nitrogen gas, high purity (≥99.99%)

5.2 Reference Standards

Compound	Source	Purity	CAS Number	M_w (g/mol)
Capsaicin	Sigma-Aldrich (Supelco CRM)	≥95%	404-86-4	305.41
Dihydrocapsaicin	Sigma-Aldrich (Supelco CRM)	≥95%	19408-84-5	307.43
Nordihydrocapsaicin	Sigma-Aldrich (analytical standard)	≥90%	28789-35-7	293.40
Homocapsaicin	Sigma-Aldrich (analytical standard)	≥85%	58493-48-4	319.44
Homodihydrocapsaicin	Sigma-Aldrich (analytical standard)	≥85%	20279-06-5	321.45

5.3 Standard Preparation

Prepare a primary stock solution of each capsaicinoid at 1000 µg/mL in methanol. Weigh 10.0 ± 0.1 mg of each CRM into separate 10 mL Class A volumetric flasks, dissolve and dilute to volume with MeOH. A mixed working stock containing all five analytes at 100 µg/mL each is prepared by combining 1.0 mL of each primary stock into a 10 mL volumetric flask and diluting to volume with MeOH. Store all standards at −20 °C in amber glass vials; discard after 6 months or upon evidence of degradation.

6 Sample Preparation

The following extraction protocol is modified from AOAC 995.03 to accommodate the elevated capsaicinoid concentrations in Scoville Splice formulations.

6.1 Drying

1. Place the representative sample aliquot in a tared aluminium weighing dish.
2. Record the wet mass (M_{wet}) to 0.1 mg.
3. Dry at 60 ± 2 °C in a forced-air oven for 48 hours, or until consecutive weighings (2 h apart) differ by ≤0.1%.
4. Record the dry mass (M_{dry}). Calculate moisture content as:

$$\text{Moisture (\%)} = [(M_{wet} - M_{dry}) / M_{wet}] \times 100$$

6.2 Grinding

1. Grind the dried material using the laboratory mill to pass a 0.5 mm sieve.
2. Homogenize by tumbling for 2 minutes.
3. Store ground material in a sealed amber glass jar at ambient temperature; use within 7 days.

6.3 Extraction

1. Weigh 1.000 ± 0.005 g of ground sample into a 50 mL polypropylene centrifuge tube.
2. Add 25.0 mL acetone. Cap tightly.
3. Vortex at maximum speed for 2 minutes.
4. Sonicate in the ultrasonic bath at 40 ± 2 °C for 30 minutes.
5. Centrifuge at 5000 × *g* for 10 minutes at ambient temperature.
6. Decant and retain the supernatant in a 100 mL round-bottom flask.
7. Repeat the extraction two additional times with 15.0 mL acetone each (vortex 2 min, sonicate 15 min, centrifuge 5000 × *g* × 10 min).
8. Pool all three supernatants.
9. Evaporate to dryness under a gentle stream of nitrogen at 40 ± 2 °C.
10. Reconstitute the residue in 10.0 mL methanol (Class A volumetric flask). Sonicate briefly to ensure complete dissolution.

11. Filter through a 0.45 µm PTFE syringe filter, discarding the first 1 mL, into an amber HPLC vial.

> **Note:** For products at the upper end of the portfolio (≥Product 08, Collateral Damage), a secondary 1:10 dilution in MeOH may be required to bring the analyte concentration within the linear range of the calibration curve. Document the dilution factor in the analytical worksheet.

7 Chromatographic Conditions

Parameter	Setting
Mode	Isocratic
Mobile Phase	ACN : H_2O : AcOH = 60 : 39.5 : 0.5 (v/v/v)
Flow Rate	1.0 mL/min
Column Temperature	25 ± 1 °C
Detection	UV absorption at 280 nm
Injection Volume	20 µL
Run Time	25 minutes
Equilibration	Minimum 10 column volumes before first injection

7.1 Expected Elution Order and Approximate Retention Times

Peak	Compound	Approximate t_R (min)
1	Nordihydrocapsaicin (NDC)	8.2
2	Capsaicin (CAP)	10.5
3	Dihydrocapsaicin (DHC)	13.1
4	Homocapsaicin (HC)	17.4
5	Homodihydrocapsaicin (HDC)	20.8

8 Beer–Lambert Law and Spectrophotometric Basis

The quantification of capsaicinoids by UV detection at 280 nm is governed by the Beer–Lambert law, which relates absorbance to the concentration of the absorbing species:

$$A = \varepsilon \cdot b \cdot c \qquad \text{Eq. 1}$$

Where:

- **A** = measured absorbance (dimensionless)
- **ε** = molar absorptivity ($L \cdot mol^{-1} \cdot cm^{-1}$)
- **b** = optical path length (cm); for a standard flow cell, b = 1.0 cm
- **c** = molar concentration of the analyte (mol/L)

For capsaicin at 280 nm, the molar absorptivity ε_{280} = 8,600 $L \cdot mol^{-1} \cdot cm^{-1}$, with a molecular weight M_W = 305.41 g/mol. The specific absorptivity (mass-concentration basis) is therefore:

$$a = \varepsilon / M_W = 8{,}600 / 305.41 = 28.16\ L \cdot g^{-1} \cdot cm^{-1} \qquad \text{Eq. 2}$$

This relationship underpins detector response linearity across the working concentration range. The minimum detectable absorbance change (baseline noise) of a modern UV detector is typically ±5 × 10^{-5} AU, establishing the theoretical floor for detection sensitivity at the chosen wavelength.

9 Calibration and Linearity

9.1 Calibration Standards

Prepare a five-point calibration curve from the mixed working stock (Section 5.3) by serial dilution in methanol:

Level	Concentration (µg/mL)	Dilution of 100 µg/mL stock
1	1	0.1 mL → 10 mL
2	5	0.5 mL → 10 mL
3	10	1.0 mL → 10 mL
4	50	5.0 mL → 10 mL
5	100	Undiluted working stock

9.2 Linear Regression Model

The calibration function is a first-order linear regression of the form:

$$y = mx + b \qquad \text{Eq. 3}$$

Where:

- **y** = detector response (peak area, mAU·s)
- **x** = analyte concentration (µg/mL)
- **m** = slope (sensitivity, mAU·s per µg/mL)
- **b** = y-intercept (should approach zero for a well-characterized system)

9.3 Acceptance Criteria

- Coefficient of determination: **$R^2 \geq 0.999$**
- y-intercept: not more than 2% of the response at the 100 µg/mL level
- Residuals: no systematic pattern; each point within ±5% of the predicted value
- Back-calculated concentrations of each calibration level within ±5% of nominal (±10% at LOQ level)

If $R^2 < 0.999$, investigate for pipetting errors, standard degradation, or detector non-linearity before repeating the calibration. A second-order (quadratic) fit is not permitted without formal method amendment and revalidation.

10 Scoville Heat Unit Calculation

10.1 Total Capsaicinoid Content

The concentration of each individual capsaicinoid in the sample is determined from the calibration curve and corrected for dilution, extraction volume, and sample mass:

$$C_i \text{ (mg/g DW)} = [x_i \times V_{final} \times D] / [M_{sample} \times 1000] \quad \text{Eq. 4}$$

Where:

- $\mathbf{x_i}$ = concentration of capsaicinoid *i* from calibration (µg/mL)
- $\mathbf{V_{final}}$ = final reconstitution volume (mL), typically 10.0 mL
- **D** = any additional dilution factor (dimensionless)
- $\mathbf{M_{sample}}$ = dry sample mass (g)

10.2 SHU Conversion

The Scoville Heat Unit rating is the weighted sum of individual capsaicinoid concentrations, each multiplied by its specific pungency coefficient (α_i). The pungency coefficients represent the SHU equivalent per unit mass of pure compound:

$$SHU = \Sigma \, (C_i \times \alpha_i \times 10^6) \quad \text{Eq. 5}$$

where C_i is expressed in g/g (mass fraction) and α_i is the pungency coefficient for each capsaicinoid:

Table 10-1. Capsaicinoid Pungency Coefficients

Capsaicinoid	Abbreviation	α_i (SHU per ppm)	Pure Compound SHU
Capsaicin	CAP	16.1	16.1×10^6
Dihydrocapsaicin	DHC	15.0	15.0×10^6
Nordihydrocapsaicin	NDC	9.2	9.2×10^6
Homodihydrocapsaicin	HDC	8.6	8.6×10^6
Homocapsaicin (Nonivamide)	NON/HC	9.1	9.1×10^6

Note: The pungency coefficients are per ASTA 21.3. Different coefficient sets exist in the literature; any change requires formal amendment and recalculation of all product specifications.

11 Limits of Detection and Quantitation

The LOD and LOQ are determined per ICH Q2(R1) using the standard deviation of the response (σ) and the slope (S) of the calibration curve:

$$LOD = 3.3 \times \sigma / S \qquad \text{Eq. 6}$$

$$LOQ = 10 \times \sigma / S \qquad \text{Eq. 7}$$

Where:

- **σ** = standard deviation of the response, determined by one of:
  - (a) Standard deviation of the y-intercepts from at least 3 independent calibration curves, or
  - (b) Standard deviation of the residuals of the regression line
- **S** = slope of the calibration curve (mAU·s per µg/mL)

11.1 Acceptance Criteria

- LOD shall be confirmed by analyzing a spiked sample at the calculated LOD concentration and verifying a signal-to-noise ratio (S/N) ≥ 3.
- LOQ shall be confirmed by demonstrating acceptable precision (RSD ≤ 10%) and accuracy (recovery 80–120%) at the LOQ concentration over 6 replicate preparations.
- LOQ must be at or below the lowest calibration level (1 µg/mL).

12 Chromatographic Resolution

Baseline resolution between adjacent capsaicinoid peaks is assessed using the USP resolution equation:

$$R_s = 2(t_{R2} - t_{R1}) / (w_1 + w_2) \qquad \text{Eq. 8}$$

Where:

- $\mathbf{t_{R1}}$, $\mathbf{t_{R2}}$ = retention times of the two adjacent peaks (min)
- $\mathbf{w_1}$, $\mathbf{w_2}$ = peak widths at base of the two peaks (min), measured by tangent construction

12.1 Acceptance Criteria

- $\mathbf{R_s \geq 1.5}$ between all adjacent capsaicinoid pairs (NDC/CAP, CAP/DHC, DHC/HC, HC/HDC)
- If $R_s < 1.5$ for any pair, verify column age, check mobile phase preparation, and adjust organic content in ±1% increments until resolution is restored.

For the critical pair (NDC/CAP), $R_s \geq 2.0$ is recommended to ensure quantitative accuracy is not compromised by partial overlap, particularly at higher capsaicinoid concentrations where peak broadening may be observed.

13 System Suitability Testing

System suitability testing is performed before each analytical run using a minimum of six (6) replicate injections of the mid-level calibration standard (10 µg/mL mixed capsaicinoid). All criteria must be met before proceeding with sample analysis.

13.1 Theoretical Plates (Column Efficiency)

$$N = 16 \times (t_R / w)^2 \qquad \text{Eq. 9}$$

Where:

- **N** = number of theoretical plates
- $\mathbf{t_R}$ = retention time of the analyte peak
- **w** = peak width at base (same time units as t_R)

13.2 Tailing Factor

$$T = w_{0.05} / (2 \times f) \qquad \text{Eq. 10}$$

where $w_{0.05}$ is the peak width at 5% of peak height, and *f* is the distance from the leading edge of the peak to the peak maximum at 5% height.

13.3 SST Acceptance Criteria Summary

Parameter	Criterion	Measured On
Theoretical plates (N)	≥ 2,000	Capsaicin peak
Tailing factor (T)	0.8 – 1.5	All capsaicinoid peaks
Injection precision (RSD)	≤ 2.0%	6 replicate injections, peak area
Resolution (R_s)	≥ 1.5	All adjacent peak pairs
Retention time RSD	≤ 1.0%	6 replicate injections, capsaicin peak

Critical: If any SST parameter fails, do not proceed with sample analysis. Diagnose the root cause (column degradation, mobile phase preparation error, detector lamp intensity, leak, etc.), correct, and re-run SST from the beginning.

14 Measurement Uncertainty Propagation

The combined standard uncertainty of the analytical result is estimated by propagating the individual uncertainty contributions from each step of the measurement process, per the EURACHEM/CITAC guide:

$$u_c = \sqrt{(u^2_{extraction} + u^2_{injection} + u^2_{calibration} + u^2_{weighing} + u^2_{volumetric})} \quad \text{Eq. 11}$$

Where the principal uncertainty components are:

Source	Symbol	Typical Relative u (%)	Estimation Method
Extraction efficiency	$u_{extraction}$	1.5–3.0	Recovery studies on spiked matrix (n ≥ 6)
Injection repeatability	$u_{injection}$	0.5–1.0	RSD of 6 replicate injections (SST data)
Calibration curve	$u_{calibration}$	1.0–2.0	Standard error of the regression, propagated to sample concentration
Weighing	$u_{weighing}$	0.05–0.1	Balance calibration certificate and repeatability
Volumetric operations	$u_{volumetric}$	0.2–0.5	Glassware tolerances (Class A certificates)

14.1 Expanded Uncertainty

$$U = k \times u_c \quad \text{Eq. 12}$$

where $k = 2$ for an approximate 95% confidence interval (assuming a normal distribution with sufficient degrees of freedom, $\nu \geq 30$). The expanded uncertainty U is reported with every analytical result and included on the Certificate of Analysis.

Typical expanded uncertainty for this method under validated conditions: **$U = 2u_c \approx$ 4–7% relative**, dominated by the extraction recovery contribution.

15 Method Validation per ICH Q2(R1)

The method shall be validated (or verified following transfer) for the following performance characteristics prior to routine use. Each parameter's acceptance criteria are specified below.

Parameter	Requirement	Acceptance Criterion
Specificity	Demonstrate no co-eluting interferences from matrix components at the retention times of all five capsaicinoids	Blank matrix injection shows no peaks > 0.1% of LOQ response at analyte t_R windows
Linearity	5-point calibration, 1–100 µg/mL, each in triplicate	$R^2 \geq 0.999$; residuals within ±5%
Range	Demonstrated from LOQ through 120% of the highest expected sample concentration	Linearity, accuracy, and precision criteria met across the range
Accuracy	Spiked recovery at 3 levels (50%, 100%, 150% of target), n = 3 per level	Mean recovery 95.0–105.0% at each level
Repeatability	6 replicate preparations from a single homogeneous sample	RSD ≤ 2.0%
Intermediate Precision	2 analysts × 2 days × 3 replicates = 12 determinations	RSD ≤ 3.0%
LOD / LOQ	Per ICH Q2(R1) as described in Section 11	LOQ ≤ 1 µg/mL; confirmed with S/N, precision, and accuracy
Robustness	Deliberate variations: column temp ±5 °C, flow ±0.1 mL/min, organic content ±2%, detection wavelength ±2 nm	SST criteria maintained; results within ±3% of nominal method

16 Per-Product QC Acceptance Criteria

Each Scoville Splice product has defined target specifications with acceptance bands for lot release. The SHU band is ±3% of the target. The capsaicin proportion range is the nominal ±2 percentage points. All values refer to dry-weight basis determinations.

Table 16-1. Product Specification and QC Acceptance Ranges

No.	Product Name	Target SHU	SHU −3%	SHU +3%	mg/g DW	CAP %	CAP % Low	CAP % High	Pass Criteria
01	Verbal Warning	3,000,000	2,910,000	3,090,000	220.0	40%	38%	42%	SHU within band AND CAP% within range
02	Kill Switch	3,600,000	3,492,000	3,708,000	261.3	44%	42%	46%	SHU within band AND CAP% within range
03	Hostile Agent	4,300,000	4,171,000	4,429,000	309.4	48%	46%	50%	SHU within band AND CAP% within range
04	Containment Breach	5,100,000	4,947,000	5,253,000	363.7	52%	50%	54%	SHU within band AND CAP% within range
05	System Shock	6,000,000	5,820,000	6,180,000	423.8	56%	54%	58%	SHU within band AND CAP% within range

No.	Product Name	Target SHU	SHU −3%	SHU +3%	mg/g DW	CAP %	CAP % Low	CAP % High	Pass Criteria
06	Venom Protocol	7,000,000	6,790,000	7,210,000	490.0	59%	57%	61%	SHU within band AND CAP% within range
07	Level 4 Pathogen	8,200,000	7,954,000	8,446,000	568.6	63%	61%	65%	SHU within band AND CAP% within range
08	Collateral Damage	9,500,000	9,215,000	9,785,000	653.1	67%	65%	69%	SHU within band AND CAP% within range
09	Extinction Protocol	11,000,000	10,670,000	11,330,000	749.2	71%	69%	73%	SHU within band AND CAP% within range
10	D.N.R.	13,000,000	12,610,000	13,390,000	877.8	75%	73%	77%	SHU within band AND CAP% within range

Note: A lot passes release testing only when *both* the SHU value falls within the ±3% band *and* the capsaicin proportion (as a percentage of total capsaicinoids) falls within the ±2 pp range. If either criterion is not met, the lot is classified OOS and the investigation procedure in Section 18 is initiated.

17 Batch Release Protocol

17.1 Sampling Plan

1. Collect a minimum of three (3) composite samples per batch, drawn from top, middle, and bottom of the blending vessel (or from three randomly selected containers for packaged product).
2. Each composite sample shall comprise at least five (5) incremental subsamples combined and homogenized.
3. Prepare and analyze each composite in duplicate, yielding a minimum of six (6) analytical results per batch.

17.2 Release Decision Tree

1. Verify that system suitability criteria (Section 13) were met for the analytical run.
2. Calculate the mean SHU and mean capsaicin proportion from all replicate determinations.
3. Confirm that the mean SHU falls within the product-specific ±3% acceptance band (Table 16-1).
4. Confirm that the mean capsaicin percentage falls within the product-specific ±2 pp range (Table 16-1).
5. Confirm that the RSD across all replicates is ≤3.0% for SHU and ≤2.0% for capsaicin proportion.
6. Verify that no individual result deviates by more than ±5% from the batch mean (outlier screening).
7. If all criteria are satisfied, the QC Analyst signs the batch record, and the QA Reviewer provides independent approval for release.

17.3 Signatures

QC Analyst — Date	QA Reviewer — Date

17.4 Retain Samples

Retain a minimum of 2× the quantity required for complete reanalysis from each released batch. Store retain samples at ≤25 °C, protected from light, for a minimum of one year beyond the product expiration date.

18 Out-of-Specification Investigation

18.1 Phase I: Laboratory Investigation

Upon identification of an OOS result, the analyst shall immediately halt testing and notify the laboratory supervisor. The Phase I investigation must be completed within 3 business days and shall include:

1. **Analyst self-check:** Review all calculations, dilutions, standard preparations, and instrument parameters for transcription or procedural errors.
2. **Data integrity review:** Examine chromatograms for integration anomalies, baseline disturbances, co-elutions, or carryover artifacts.
3. **System suitability verification:** Confirm that all SST parameters passed for the run in question.
4. **Instrument check:** Verify detector lamp energy, mobile phase composition, injection volume precision, and column pressure.

If a definitive assignable laboratory cause is identified (e.g., incorrect dilution, pipetting error with documented evidence), the original result is invalidated and the sample is re-prepared and re-analyzed.

18.2 Phase II: Full-Scale Investigation

If no assignable laboratory cause is found in Phase I, a full-scale OOS investigation is initiated under the direction of Quality Assurance:

1. Re-prepare and re-analyze the original sample in duplicate by the original analyst.
2. Prepare and analyze a second composite sample by a second qualified analyst on a different instrument (if available).
3. Review batch manufacturing records for deviations in raw material weighing, blending time, blending speed, or environmental conditions.
4. Evaluate trending data from previous batches to identify out-of-trend patterns.
5. If the reanalysis results conform to specification and the original result is attributable to sample preparation error, the batch may be released with documented justification.
6. If reanalysis confirms the OOS result, the batch is rejected and segregated pending a manufacturing investigation and potential rework or disposition.

18.3 Documentation

All OOS investigations shall be documented on the OOS Investigation Form (QA-F-OOS-001), including root cause determination, corrective action, preventive action (CAPA), and final disposition. The completed form requires signatures from the investigating analyst, laboratory supervisor, and Quality Assurance Director.

19 Safety and Waste Disposal

19.1 Personal Protective Equipment

- Nitrile gloves (double-gloving recommended when handling concentrated capsaicinoid standards)
- Safety glasses with side shields or chemical splash goggles
- Laboratory coat
- Perform all grinding and weighing of capsaicinoid-containing materials in a fume hood

19.2 First Aid

- **Skin contact:** Remove contaminated clothing. Wash affected area with vegetable oil or isopropanol, then with soap and water. Do not use hot water.
- **Eye contact:** Flush immediately with copious water for at least 15 minutes. Seek medical attention.
- **Inhalation:** Move to fresh air. If respiratory irritation persists, seek medical attention.

19.3 Waste Disposal

- Organic solvent waste (acetone, MeOH, ACN, mobile phase): collect in designated halogenated/non-halogenated organic waste containers per institutional EH&S protocols.
- Capsaicinoid-contaminated solid waste: collect separately and dispose as chemical waste.
- HPLC column waste: consult manufacturer for disposal or recycling instructions.

20 Revision History

Version	Date	Author	Description of Change
1.0	October 2026	Analytical Chemistry Group	Initial release. Establishes validated RP-HPLC method for capsaicinoid quantification and SHU determination across all 10 Scoville Splice products.